3 laboratory task

We will examine the data describing the activities of the parcel delivery and logistics company. The given file contains partially structured data on stops of shippers, which can be described by the following parameters: "marsrutas","sustojimo data","sandelio id","Firma","Marsruto tipas","Masinos tipas","sustojimo tipas","sustojimo savaites diena","laikas","Sustojimo numeris","siuntu skaicius","svoris","svorio grupe","geografine zona","pasto kodas", "Aptarnavimo grupe","tipas","Laikas iki sustojimo","Laikas po sustojimo", "Uzkrovimo tipas","Ar reikalingos paletes","Laukia","Sustojimo klientu skaicius","sustojimo klientu sarasas","kaina procentas","kaina vienetais".

In addition, there is data on routes on different days of the week (i.e. data grouped by the following parameter pair: "marsrutas", "sustojimo data") with the following parameters: "marsrutas", "sustojimo data","M", "BendrasAtstumas","BendrasSvoris","BendrasLaikas","BendraKaina".

With Apache Spark and DataFrame data structure, perform on of the following tasks:

1. Investigate the linear relationship between the "BendraKaina" parameter and the "siuntu skaicius" parameter (aggregate by route and date, apply the sum operation) when only one data with the same value "Masinos tipas" is considered. Apply linear regression during the analysis.

2. Investigate the linear relationship between the "BendraKaina" parameter and the "svoris" parameter (aggregate by route and date, apply the sum operation), when only one data with the same value "Masinos tipas" is considered. Apply linear regression during the analysis.

3. Investigate the linear relationship between the "BendrasLaikas" parameter and the "siuntu skaicius" parameter (aggregate by route and date, apply the sum operation) when only one data with the same value "Masinos tipas" is considered. Apply linear regression during the analysis.

4. Investigate the linear relationship between the "BendrasLaikas" parameter and the "svoris" parameter (aggregate by route and date, apply the sum operation) when only one data with the same value "Masinos tipas" is considered. Apply linear regression during the analysis.

Calculate the task number according to the formula $((n-1) \mod 4)+1$, where n – is your number in the group list.

Make a work report. In the report, describe the task according to your option; present the decisions in the form of a pseudocode, comment on the result obtained.

- A description of the linear regression of the Spark ML can be found in the official documentation: https://spark.apache.org/docs/latest/ml-classification-regression.html#linear-regression